2. **Say True or False.**

   (i) Depreciation value is calculated by the formula, $P\left(1-\frac{r}{100}\right)^n$.

   (ii) If the present population of a city is P and it increases at the rate of $r\%$ p.a, then the population n years ago would be $P\left(1+\frac{r}{100}\right)^n$.

   (iii) The present value of a machine is ₹16800. It depreciates at 25% p.a. Its worth after 2 years is ₹9450.

   (iv) The time taken for ₹1000 to become ₹1331 at 20% p.a, compounded annually is 3 years.

   (v) The compound interest on ₹16000 for 9 months at 20% p.a, compounded quarterly is ₹2522.

3. Find the compound interest on ₹3200 at 2.5% p.a for 2 years, compounded annually.

4. Find the compound interest for $2\frac{1}{2}$ years on ₹4000 at 10% p.a, if the interest is compounded yearly.

5. A principal becomes ₹2028 in 2 years at 4% p.a compound interest. Find the principal.

6. In how many years will ₹3375 become ₹4096 at $13\frac{1}{3}$ % p.a if the interest is compounded half-yearly?

7. Find the C.I on ₹15000 for 3 years if the rates of interest are 15%, 20% and 25% for the I, II and III years respectively.

8. Find the difference between C.I and S.I on ₹5000 for 1 year at 2% p.a, if the interest is compounded half yearly.

9. Find the rate of interest if the difference between C.I and S.I on ₹8000 compounded annually for 2 years is ₹20.

10. Find the principal if the difference between C.I and S.I on it at 15% p.a for 3 years is ₹1134.

Objective Type Questions

11. The number of conversion periods in a year, if the interest on a principal is compounded every two months is_____.

    (A) 2 (B) 4 (C) 6 (D) 12

12. The time taken for ₹4400 to become ₹4851 at 10%, compounded half yearly is _____.

    (A) 6 months (B) 1 year (C) 1 years (D) 2 years

13. The cost of a machine is ₹18000 and it depreciates at $16\frac{2}{3}$% annually. Its value after 2 years will be_____.

    (A) ₹12000 (B) ₹12500 (C) ₹15000 (D) ₹16500

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